

Junho Peter Whang

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Citizenship: Canada

Academic Positions

Seoul National University, Seoul, Korea

Aug. 2021–present

Assistant Professor, Department of Mathematical Sciences

Princeton University, Princeton, NJ, USA

Sep. 2025–Jun. 2026

Visiting Fellow, Department of Mathematics

Massachusetts Institute of Technology, Cambridge, MA, USA

Jul. 2018–Jun. 2021

C.L.E. Moore Instructor

Education

Princeton University, Princeton, NJ, USA

2018

Ph.D. in Mathematics. Advisor: Peter Sarnak.

Thesis: *Diophantine Analysis on Moduli of Local Systems*.

Queen's University, Kingston, ON, Canada

2013

B.Sc. in Mathematics, Honours with Distinction.

Research Interests

Number theory (Diophantine analysis), arithmetic geometry, topology, and dynamics.

Publications and Preprints

1. *On indefinite integral ternary quadratic forms*.
With Alexander Gamburd, Amit Ghosh, and Peter Sarnak. Preprint, arXiv:2603.05849.
2. *Degree gaps in algebraic interpolations*.
Bull. Korean Math. Soc. **63** (2026), no. 1, 227–234.
3. *A reciprocity law on the one-holed torus*.
J. Korean Math. Soc. **62** (2025), no. 6, 1525–1538.
4. *On finite categories of algebraic varieties*.
J. Algebra **681** (2025), 206–217.
5. *Stokes matrices and exceptional isomorphisms*.
With Yu-Wei Fan. Math. Ann. **390** (2024), 4041–4086.
6. *On periodic orbits of polynomial maps*.
Preprint, arXiv:2305.13529.
7. *Surface group representations in $\mathrm{SL}_2(\mathbb{C})$ with finite mapping class orbits*.
With Indranil Biswas, Subhojoy Gupta, and Mahan Mj. Geom. Topol. **26** (2022), no. 2, 679–719.
8. *The rank two p -curvature conjecture on generic curves*.
With Anand Patel and Ananth Shankar. Adv. Math. **386** (2021).
9. *Arithmetic of curves on moduli of local systems*.
Algebra Number Theory **14** (2020), no. 10, 2575–2605.
10. *Nonlinear descent on moduli of local systems*.
Israel J. Math. **240** (2020), 935–1004.

11. *Global geometry on moduli of local systems for surfaces with boundary.*
Compos. Math. **156** (2020), no. 8, 1517–1559.
12. *The uncertainty principle and a generalization of a theorem of Tao.*
With M. Ram Murty. Linear Algebra Appl. **437** (2012), no. 1, 214–220.
13. *Another proof of the infinitude of the prime numbers.*
Amer. Math. Monthly **117** (2010), no. 2, 181.

Grants and Research Funding

Samsung Science and Technology Foundation 2022–2027

Analysis of Diophantine Problems with Nonlinear Symmetry. SSTF-BA2201-03.

Principal Investigator. Total project award: KRW 755 million.

National Research Foundation of Korea 2025–2028

Dynamics and Arithmetic Structures on Manifolds (DASOM). RS-2025-02293115.

Co-investigator. Total project award: KRW 500 million.

Selected Honors and Awards

Teaching and Learning Award, Massachusetts Institute of Technology 2021

Bershadsky Family Fellowship in Mathematics, Princeton University 2017–2018

NSERC Postgraduate Scholarship PGS D 2014–2017

Research Supervision

Ph.D. students

- Seong Youn Kim, 2022–present; co-advised with Gye-Seon Lee.
- Young Joon Ley, 2022–present.
- Eungchan Kim, 2025–present.
- Jihye Oh, 2025–present.

Master’s students

- Juhan Kim, M.S. 2023.
Thesis: *Compactification of Character Varieties via Triangulation on Surfaces.*
- Eunju Shin, M.S. 2024.
Thesis: *Nonlinear Reduction for Two Types of Generalized Markoff Equations.*
- Jihye Oh, M.S. 2025.
Thesis: *Decidability of Mutation Equivalence for Three-Vertex Quivers.*

Teaching

Seoul National University 2021–present

Graduate: Algebra, Algebraic Geometry, Commutative Algebra, Diophantine Geometry.

Advanced Topics in Number Theory.

Undergraduate: Linear Algebra, Introduction to Diophantine Analysis, Analytic Number Theory.

Massachusetts Institute of Technology 2018–2021

Introduction to Arithmetic Geometry (18.782).